Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 2 · SLO 1 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 2 : SLO â€“ 1 | Array Stack â€” Advantages, Disadvantages, Applications

I. Advantages of Array Implementation

- Stack implementation using arrays is EASY (simple) to implement.
- Accessing elements in an array is FAST due to direct indexing.
- Stack operations like push and pop take O(1) (constant) time in array implementation.
- Arrays provide STATIC memory allocation at compile time.
- Array-based stacks have CONTIGUOUS (cache-friendly) memory access, improving performance.
- There is no overhead of memory allocation and DEALLOCATION (pointer management) in arrays.
- Array-based stack implementation avoids the extra memory used for POINTERS in linked lists.
- It is easier to IMPLEMENT (code) a stack implemented using arrays in programming.
- Array implementation of stack is EFFICIENT in terms of memory for small, fixed-size stacks.
- Stack overflow in array implementation is EASY to detect.

II. Disadvantages of Array Implementation

- Stack implemented using arrays has a fixed MAXIMUM size.
- If the stack size is too small, it may lead to OVERFLOW.
- If the stack size is too large, it may lead to WASTAGE of memory.
- Arrays use STATIC memory allocation, which is less flexible.
- It is difficult to RESIZE the size of a stack implemented using arrays at runtime.
- Stack operations using arrays are not memory EFFICIENT for large data.
- Inserting or removing elements from the stack may require checking for OVERFLOW or underflow.
- Reallocation of array memory is COSTLY and not always supported in some languages.
- Array implementation is not suitable when the maximum stack size is UNKNOWN (unpredictable).
- Stack implemented using arrays may not efficiently use MEMORY in dynamic scenarios.

III. Applications of Stacks

- Stacks are used in EXPRESSION evaluation and conversion.
- Infix, postfix, and prefix expressions are evaluated using STACKS.
- Stacks help in checking balanced PARENTHESES in an expression.
- Stack is used to perform DEPTH-FIRST traversal in trees.
- The function call mechanism uses a CALL stack.
- Stacks are used in BACKTRACKING algorithms for backtracking.
- Undo and redo operations in editors are managed using STACKS.
- Stack is used in reversing a STRING (list).
- Browser history navigation (back and forward) uses the concept of STACKS.
- Stacks are used in memory management and RECURSION handling in programs.